

DR. NGOZI MARGARET OGUGUAH, PhD

Chief Research Officer | Marine & Fisheries Scientist | Science–Policy Practitioner

Nigerian Institute for Oceanography and Marine Research (NIOMR), Lagos, Nigeria

Address: 3 Wilmot Point Road, Victoria Island, Lagos, Nigeria

Email: ngozimoguguah@yahoo.com | **Phone:** +234 8032642262

ORCID: [0000-0002-8486-0440](https://orcid.org/0000-0002-8486-0440) | **LinkedIn:** [linkedin.com/in/ngozi-oguguah-72252426](https://www.linkedin.com/in/ngozi-oguguah-72252426) | **ResearcherID:** researchid.co/ngozimoguguah

Academic Profiles: Google Scholar: [profile](#) | Scopus: [Author ID 57195291552](#) | ResearchGate: [profile](#)

Marine scientist, fisheries biologist, science-policy practitioner, and internationally recognized leader in marine pollution research and ocean governance, with more than two decades of experience advancing aquatic ecosystem research, fisheries management, plastic pollution monitoring, and evidence-informed environmental policy in West Africa and globally.

Currently serving as **Chief Research Officer** at the Nigerian Institute for Oceanography and Marine Research (NIOMR), she leads interdisciplinary research that integrates field ecology, remote sensing, drones, GIS, artificial intelligence, and stakeholder engagement to address complex environmental challenges.

Recognized internationally for pioneering drone- and machine learning-based monitoring of plastic pollution in Nigeria and for translating scientific evidence into environmental policy. Passionate about strengthening the science-policy interface, mentoring emerging scientists, advancing equitable ocean governance, and communicating science through storytelling, documentaries, and public engagement.

Selected impact highlights: pioneered drone- and AI-enabled plastic pollution monitoring in Nigeria; mentored more than 250 early-career scientists and students across Africa; contributed to UN Ocean Decade, BBNJ, ASLO, COESSING, and Ocean Sciences Meeting leadership platforms; and produced public-facing documentaries that translate marine science for wider audiences.

Career Highlights

- Chief Research Officer, Nigerian Institute for Oceanography and Marine Research (NIOMR), since 2023
- 20+ years advancing marine and fisheries science, pollution monitoring, and evidence-based ocean policy
- Pioneered drone- and AI-enabled plastic pollution monitoring in Nigeria
- Mentored more than 250 early-career scientists and students across Africa
- ASLO Fellow; first African elected to the ASLO Board of Directors
- Co-Chair, Ocean Sciences Meeting (OSM) 2026 Planning Committee
- Principal Investigator, Royal Academy of Engineering Seed Grant — drone- and machine-learning-based plastic pollution mapping in Lagos
- Executive Producer and Director, Plastic Peril documentary on coastal plastic pollution in Nigeria
- Contributed to the UN Ocean Decade for Africa publication and served as civil society advocate at BBNJ IGC-5, UN Headquarters
- Co-Organizer and Faculty, Coastal Ocean Environment Summer School in Nigeria and Ghana (COESSING)

Research Metrics: [Add total peer-reviewed publications, citation count, and h-index from Google Scholar/Scopus]

Research Areas

- **Marine and plastic pollution monitoring**, including microplastics and nanoplastics
- **Fisheries biology and aquatic biodiversity conservation**
- **Coastal ecosystem management and marine hazard forecasting**
- **Science–policy interface and ocean governance**, including BBNJ and blue economy priorities
- **Climate change adaptation and coastal resilience**
- **GIS, remote sensing and artificial intelligence** for environmental monitoring
- **Environmental toxicology and human health risk assessment**

Key contributions: advanced technology-enabled marine pollution monitoring in Nigeria; strengthened science-policy translation through government and international platforms; expanded ocean-science capacity through training, mentorship, and COESSING; and broadened public understanding of marine sustainability through documentary production and media engagement.

Honours and Awards

- **Senior Fellow**, Coastal Institute, University of Rhode Island, USA
- Black in Marine Science Tidal Wave Fellow (2023)
- **Association for the Sciences of Limnology and Oceanography Fellow** (2023)
- **Beahrs Environmental Leadership Fellow**, University of California, Berkeley, USA (2021/2023)
- **Scripps-Rady Ocean Plastic Pollution Challenge Fellow**, UC San Diego (2021)
- **MarXiv Global Ambassador** (2017)
- **ASLO Global Outreach Grant Winner** (2016)
- **African Women in Agricultural Research and Development Fellow** (2014)
- **Norman E. Borlaug African Women in Science Fellow**, USDA/University of Florida (2010)

Education

- **Ph.D. in Zoology (Fisheries Biology)**, University of Nigeria, Nsukka, Nigeria, 2021. Dissertation: *Heavy metal concentration in water, sediment and some commercially important fish species of Lagos Lagoon and its human implications.*
- **M.Sc. in Zoology (Fisheries Biology)**, University of Nigeria, Nsukka, Nigeria, 2004–2006. Thesis: *Production parameters of *Heterobranchus bidorsalis* and *Clarias gariepinus* hybrid juveniles stocked at different stocking densities in varied culture tanks.*
- **B.Sc. in Zoology/Biochemistry**, University of Nigeria, Nsukka, Nigeria, 1990–1995.

International Leadership and Academic Service

- **Co-Chair, Planning Committee** | Ocean Sciences Meeting (OSM) 2026
- **Vice Co-Chair, Planning Committee** | Ocean Sciences Meeting (OSM) 2024
- **Planning Committee Member** | Ocean Sciences Meeting (OSM) 2020
- **Civil Society and Community Advocate** | Intergovernmental Conference on Marine Biodiversity of Areas Beyond National Jurisdiction (BBNJ IGC-5), United Nations Headquarters, New York, August 2022

- **Co-Organizer and Faculty** | Coastal Ocean Environment Summer School in Nigeria and Ghana (COESSING), 2022 and 2024; supported regional capacity sharing in coastal oceanography through teaching, mentoring, and international collaboration.
- **Early Career Ocean Professional Co-Lead** | IOC-UNESCO Evolving and Sustaining Ocean Best Practices Workshop IV, 2020
- **Board of Directors Member** | Society for Women in Marine Science
- **Executive Committee Member** | Ocean Knowledge Action Network
- Steering Committee Member, Science and Risk Communication Interest Group, SETAC (2019)
- ASLO Student Board Member (2017) — first African elected to the ASLO Board of Directors, marking an early international leadership milestone in limnology and oceanography.

Academic and Research Appointments

Chief Research Officer

Nigerian Institute for Oceanography and Marine Research (NIOMR), Lagos, Nigeria | January 2023–Present

- Direct multi-disciplinary research projects assessing land-based and marine pollution, plastic waste trajectories, and ecotoxicology in West African waters.
- Lead national initiatives on climate change adaptation in fisheries, aquaculture, and finfish biodiversity conservation.
- Deliver postgraduate training and supervise undergraduate researchers via the SIWES industrial internship framework.

Principal Research Officer

Nigerian Institute for Oceanography and Marine Research (NIOMR), Lagos, Nigeria | January 2014–December 2022

- Managed research teams investigating heavy metal, nutrient, and microplastic pollution across the Lagos Lagoon ecosystem.
- Facilitated science-policy translation through evidence synthesis for state and national ministries.

Senior Research Officer

Nigerian Institute for Oceanography and Marine Research (NIOMR), Lagos, Nigeria | December 2011–December 2013

- Coordinated field sampling, water quality indexing, and coastal erosion studies.

Research Officer

Nigerian Institute for Oceanography and Marine Research (NIOMR), Lagos, Nigeria | 2008–2011

- Conducted bioassays, fish stock monitoring aboard research vessel MV Sussana, and baseline water chemistry profiling.

Research Leadership and Impact

- **Pioneered drone-AI plastic monitoring:** Directed a project leveraging low-altitude photogrammetry drones combined with machine learning to quantify plastic pollution hotspots along urban coastlines.
- **Advanced regional ecotoxicology benchmarks:** Co-led research generating evidence on microplastic ingestion by commercially vital fish species across the eastern equatorial Atlantic Ocean.
- **Strengthened policy translation and advisory practice:** Embedded within the Lagos State Ministry of the Environment and Water Resources through the EUEPiN project to align scientific monitoring data with state policy formulation.
- **Expanded science communication through documentary filmmaking:** Produced *Plastic Peril* and *Women in Seafood* to translate marine science into compelling public narratives.

Research Grants and Funded Projects

- **Royal Academy of Engineering, UK — Seed Grant** | Principal Investigator | *Mapping Plastic Pollution Hotspots in Lagos Using Drones and Machine Learning*. [Add funding amount, duration, and 1–2 outcome bullets.]
- **Royal Academy of Engineering, UK — Frontiers Grant** | Co-Investigator | *Sustainable Ship Breaking Process for Cleaner Environment and Better Well-being*. [Add funding amount, duration, and 1–2 outcome bullets.]
- **Global Challenges Research Fund Project** | Co-Investigator | *Micro and Nanoscale Plastics in Coastal Ghana: Implications for Ecosystem Functioning and Human Health*. [Add funding amount, duration, and 1–2 outcome bullets.]
- **ASLO Global Outreach Grant** | Lead Awardee | *Limnological and Oceanographic Practical Capacity Transfer to Undergraduates*. [Add funding amount, duration, and 1–2 outcome bullets.]
- **Scripps-Rady Ocean Plastic Challenge Support** | Fellow and White Paper Lead | *Evaluating Efficacy of Plastic Pollution Solutions in the Southern California Bight*. [Add funding amount, duration, and 1–2 outcome bullets.]

Science–Policy Leadership

- **Policy Attachment Fellow** | Lagos State Ministry of the Environment and Water Resources, Evidence Use in Environmental Policymaking in Nigeria project. Served as an expert advisor providing data-driven recommendations on coastal pollution control.
- **UN Ocean Decade contribution:** Co-authored the UN Ocean Decade for Africa publication, *The Science We Need for the Ocean We Want in Africa*, in partnership with WIOMSA and IOC-UNESCO, contributing to a continental science agenda for ocean sustainability.
- **High Seas Treaty engagement:** Served as community advocate at the United Nations Headquarters during BBNJ IGC-5, supporting global governance for marine biodiversity in international waters.

Technology, Innovation and Inventions

- **Computer-Vision Coastal Monitoring Workflow:** Co-developed an open-source workflow using drone imagery and YOLO object detection models for automatic marine debris classification.

- **Appropriate Aquaculture Aeration Technology:** Designed low-cost aquaculture tank aeration mechanisms utilizing repurposed medical intravenous drip sets, syringe tubes, and stone diffusers to support resource-limited fish farmers.

Selected Publications

Peer-Reviewed Journal Articles

- **Oguguah, N.M.** (2025). Book review: *Water Chemistry: The Chemical Processes and Composition of Natural and Engineered Aquatic Systems* (2nd ed., P.L. Brezonik & W.A. Arnold). *Limnology and Oceanography Bulletin*, 34, 116–117.
- **Oguguah, N.M.** (2025). Autobiographical sketches. *Oceanography*, 38(1), 116–128.
- Arbic, B.K., Adjetei, J., Agyekumhene, A., Akinwunmi, M.F., Akita, L.G., Anderson, L., Ansong, J.K., Addo, K.A., Asamoah, E.K., Awe, O.O., Buckingham, C., Collier, J., Cotel, A.J., Damoah, R., Elegbeleye, O.W., Farneti, R., Foster-Martinez, M., Frank, C., Howden, S., Idowu, R.T., Johnson, W.M., Kaiser, D., Knoop, P., Lamptey, A.M., Lawal-Are, A., Lucas, A.J., Mahu, E., Martin, P., Menemenlis, D., Nubi, O.A., Nyadjro, E.S., Nyarko, E., **Oguguah, N.**, Oikonomou, A., Oladipo, M., Osborne, T., Quarcoo, R.K., Quaye, D.T., Saba, A.O., Simon, A.C., Sogbanmu, T.O., Tsei, S., & Vagenas, G. (2025). The Coastal Ocean Environment Summer School in Nigeria and Ghana: The value of long-term, sustained capacity sharing. *Oceanography*, 38(1).
- Vigness-Raposa, K.J., Van Uffelen, L.J., Oladipo, M.O., Asamoah, E.K., Saba, A.O., **Oguguah, N.M.**, Lawal-Are, A.O., & Becker, K.M. (2024). Underwater acoustics for all: Expanding capacity with education and low-cost sensors. *Oceanography*. <https://doi.org/10.5670/oceanog.2025.107>
- Roman, R., Kostianaia, E., Bowes, A., Satterthwaite, E., Savage, C., Ilosvay, X.E.E., Fox, N., Boujmil, I., Nocito, E., Santamaria, S.M., Maggioni, T., Ndah, A., Onwubiko, C., Higa, K., Juma, G.A., Mwangi, P.N., Offei-Darko, L., Khatib, L., Elegbede, I., **Oguguah, N.**, Elkhoully, H.A., & Nwajiaku, L. (2024). Building bridges for ocean sustainability: The evolution and impact of the Early Career Ocean Professional Programme. *Marine Technology Society Journal*, 58. <https://doi.org/10.4031/MTSJ.58.1.4>
- Ngwa, C.H., Mpofu, L., Patricia, T., Libwea, J.N., Obiora, R.U., Keinamma, M., Ines, D.A., Ngo, N.V., Oneke, E.H., Bessong, C.E., **Oguguah, N.M.**, Kah, E., Wirsiy, F.S., & Sobh, E. (2024). Prevalence, risk factors and management of common mental health disorders in Cameroon: A systematic review. *BMJ Public Health*, 2, e000224. <https://doi.org/10.1136/bmjph-2023-000224>
- Gushulak, C.A.C., Bodmer, P., Carvalho, C.R., Gladstone-Gallagher, R.V., Lee, Y.P., Lowman, H.E., **Oguguah, N.M.**, Meinikmann, K., Rii, Y.M., Rodríguez-Cardona, B.M., & Bizic, M. (2023). The silent mental health and well-being crisis of early career researchers in aquatic sciences. *Limnology and Oceanography Bulletin*. <https://doi.org/10.1002/lob.10539>
- Mahu, E., Datsomor, W.G., Folorunsho, R., Fisayo, J., Crane, R., Marchant, R., Montford, J., Boateng, M.C., Oti, M.E., **Oguguah, N.M.**, & Gordon, C. (2022). Human health risk and food safety implications of microplastic consumption by fish from coastal waters of the eastern equatorial Atlantic Ocean. *Food Control*. <https://doi.org/10.1016/j.foodcont.2022.109503>
- Adekola, O., Namawejje, H., **Oguguah, N.**, Onyegbulam, L., Nweze, V., Abasilim, A., Ikegwu, O., & Mulema, A. (2022). How COVID-19 has affected research productivity in Africa: Lessons for the future. *African Geographical Review*. <https://doi.org/10.1080/19376812.2022.2063142>
- Rizvi, M.J., Islam, M.R., Adekola, O., & **Oguguah, N.M.** (2020). A sustainable shipbreaking approach for cleaner environment and better wellbeing. *Journal of Cleaner Production*, 270, 122522. <https://doi.org/10.1016/j.jclepro.2020.122522>

- Moore, T., & **Oguguah, N.** (2018). Student activities in Victoria, ASLO summer meeting 2018. *Limnology and Oceanography Bulletin*. <https://doi.org/10.1002/lob.10270>
- Elser, J., Moore, T., & **Oguguah, N.** (2018). How to become a member of the ASLO Board in three easy steps. *Limnology and Oceanography Bulletin*. <https://doi.org/10.1002/lob.10250>
- **Oguguah, N.M.**, Wilfred-Ekpirikpo, P.C., & Ayorinde, A.O. (2017). Practical skills transfer to undergraduate students: Physicochemical and biological parameters as indicators of Lagos Lagoon health status. *Limnology and Oceanography Bulletin*, 26(3), 73–75. <https://doi.org/10.1002/lob.10195>
- **Oguguah, N.M.**, & Ikegwu, M.O.J. (2017). Concentration and human health implications of trace metals in fish of economic importance in Lagos Lagoon. *Journal of Health & Pollution*, 7(13). <https://doi.org/10.5696/2156-9614-7-13.66>
- Iloba, K.I., **Oguguah, N.M.**, & Nwadukwe, F.O. (2016). Hydrogen ion (pH), ammonia, dissolved oxygen and nitrite concentrations in catfish water recirculatory and semi-intensive static-water systems. *Journal of Aquatic Sciences*, 31(2B), 417–424.
- **Oguguah, N.M.**, Nwadukwe, F., Atama, C.I., Chidobem, J.I., & Eyo, J.E. (2011). Growth performance of hybrid catfish (*Heterobranchus bidorsalis* ♀ × *Clarias gariepinus* ♂) at various stocking densities in varied culture tanks. *Animal Research International*, 8(3), 1419–1430.
- Oshisanya, K.I., Unyimadu, J.P., Shelle, R.O.D., Nubi, O.A., Ladigbolu, I.A., **Oguguah, N.M.**, Olumodeji, O.O., Adeleye, A.O., & Fashade, A.O. (2011). Seasonal variation of heavy metals in sediment and water of Lagos Lagoon. *Journal of American Science*, 7(3), 385–387.
- **Oguguah, N.M.**, & Eyo, J.E. (2007). Nigerian finfish feed technology. *Journal of Research in Bioscience*, 3(1), 23–39.

Books, Chapters and Technical White Papers

- Ozor, N., **Oguguah, N.M.**, & Ugwuoke, M. (2011). Sharing the Lagos mega-city experience in the integrated management of sea-level rise and flooding. In *Resilient Cities* (Local Sustainability, Vol. 1, Part 4, pp. 319–327). Springer. https://doi.org/10.1007/978-94-007-0785-6_33
- WIOMSA / IOC-UNESCO. (2022). *United Nations Ocean Decade for Africa: The science we need for the ocean we want in Africa*. WIOMSA, Zanzibar, Tanzania.
- Wiggin, K., Phipps, K., Duncan, C., Fischbach, S., **Oguguah, N.**, Robuck, A., Torres, T., Lieberg Clark, P., Jackson, E., & Stokes, D. (2021). *Evaluating efficacy of plastic pollution solutions in the Southern California Bight*. White paper of the 2021 Scripps-Rady Ocean Plastic Pollution Challenge.

Keynote and Conference Presentations

Selected International Conferences

- **Oguguah, N.**, Sogbanmu, T., Chukwudebelu, J., Lasisi, A., Abimbola, A., Wan, J., Udeinya, K., Oguguah, D.U., & Biaoku, D.O. (2024). *Mapping plastic pollution hotspots in Lagos: Integrating drone technology and machine learning for coastal ecosystem monitoring*. ASLO Meeting, Madison, Wisconsin, USA.
- **Oguguah, N.M.**, & Eyo, J.E. (2024). *Seasonal and spatial variations in physicochemical and heavy metal content of water in the Lagos Lagoon, Nigeria*. Ocean Sciences Meeting, New Orleans, Louisiana, USA.
- **Oguguah, N.M.**, & Nnodim, N.N. (2023). *Culture and traditions of Nigerian coastal dwellers and influence of the environment*. ASLO Aquatic Sciences Meeting, Palma de Mallorca, Spain.

- **Oguguah, N.M.**, Ikegwu, O.J., & Eyo, J. (2019). *Monitoring cadmium and lead in selected fish from Lagos Lagoon, Nigeria: Levels and human health assessment*. OceanVisions 2019, Georgia Tech, Atlanta, Georgia, USA.
- **Oguguah, N.M.** (2016). *Heavy metal (Pb, Cd, Fe, Zn and Mn) levels in sediments from Nigerian coastal waters*. Poster presentation, Ocean Sciences Meeting, New Orleans, Louisiana, USA.
- **Oguguah, N.M.**, & Olakolu, F.C. (2014). *Heavy metal levels in shrimp by-catch species from industrial shrimp trawl fishery in Nigerian coastal waters*. 2nd International Conference on Oceanography, Las Vegas, Nevada, USA.
- **Oguguah, N.M.**, Renner, K.O., & Oshisanya, K.I. (2011). *Composition of marine debris in Nigerian coastal waters*. 5th International Marine Debris Conference, Honolulu, Hawaii, USA.

Regional and Invited Science–Policy Talks

- **Oguguah, N.M.**, Sogbanmu, T.O., & Abasilim, A.N. (2020). *Science and policy interface by evidence producers and users: Outcomes of a secondment to an environmental policymaking institution in Nigeria*. 8th NiWARD Conference, Ilorin, Nigeria.
- **Oguguah, N.M.**, & Renner, K.O. (2010). *The role of investment in research and development in Africa: Key to sustainable growth*. African Technology Policy Studies Annual Conference, Cairo, Egypt.

Science Communication and Public Engagement

Documentary Productions

- ***Plastic Peril*** — Executive Producer and Director; documentary highlighting coastal plastic pollution in Nigeria.
- ***Women in Seafood*** — Producer; multimedia project documenting female labour, leadership, and socioeconomic challenges in artisan fisheries.
- ***Oceans are Life*** — Producer/Collaborator; short documentary produced in partnership with Greenpeace advocating for High Seas protection.

Public Engagement and Media Outreach

- **BIMS Reads Host:** Moderated an international Q&A session with author Alexis Pauline Gumbs on *Undrowned: Black Feminist Lessons from Marine Mammals*.
- **STEM Girls Talk Show:** Featured guest speaker discussing career pathways in ocean science for young women.
- **Broadcast Interviews:** Featured on Clevenard Radio and All-Atlantic Ocean Research Alliance campaigns focused on plastic-free oceans and marine biodiversity conservation.

Teaching, Training and Mentorship

Professional Mentorship Roles

- **Sustainable Ocean Alliance:** Fellowship Mentor (2025).
- **Black in Marine Science:** Tidal Wave Peer Mentor (2024–Present).
- **The Project Kuongoza Initiative / STEMi Africa:** Senior Mentor.
- **Commonwealth Mentorship Program:** Young Women’s Mentor (2019–Present).
- **Global STEM Alliance / New York Academy of Sciences:** Mentor, *1000 Girls, 1000 Futures* Program (2019–Present).

- **Cherie Blair Foundation:** Business and Mentorship Fellow for Women in Science (2016–Present).
- **African Changemakers Initiative:** Mentorship Lead (2017–Present).

Institutional Teaching and Student Training

- **SIWES Student Training:** Direct practical training and laboratory instruction for undergraduate interns in marine chemistry, ecology, and data analysis at NIOMR.
- **Research Mentorship:** Mentored more than 250 early-career scientists, graduate students, and undergraduate researchers across Africa.

Technical Skills and Competencies

- **Geospatial and Remote Sensing:** ArcGIS, QGIS, Google Earth Engine and low-altitude drone photogrammetry.
- **Data Science and Artificial Intelligence:** Machine learning, YOLO object detection, Python, R, SPSS, and quantitative and qualitative data analysis.
- **Laboratory and Field Techniques:** Heavy metal ecotoxicology, water chemistry analysis, microplastic extraction and fish stock surveying.
- **Policy and Communication:** Science-policy translation, evidence-based advisory practice, grant writing, public speaking, documentary production and stakeholder facilitation.

Professional Memberships

- Association for the Sciences of Limnology and Oceanography (ASLO)
- The Oceanography Society (TOS)
- Black in Marine Science (BIMS)
- Early Career Ocean Professional (ECOP) Africa
- Society for Women in Marine Science (SWMS)
- Organization for Women in Science for the Developing World (OWSD)
- Society of Environmental Toxicology and Chemistry (SETAC)
- The Marine Biological Association (MBA)
- World Aquaculture Society (WAS)
- Nigerian Environmental Society (NES)
- Nigerian Women in Agricultural Research for Development (NiWARD)